KUKU AI GENERALIST ASSIGNMENT | PROBLEM STATEMENT 3

SceneLock

AI pre-generation quality assurance for microdrama shots | Aditya Gautam | Roll 230106003

The problem and the person affected

An AI video artist converts a screenplay beat into a prompt for video generation. A changed costume, missing story action, or reversed handover can survive into a plausible-looking but wrong shot. SceneLock targets **one artist, one shot, one pre-generation QA decision**; it does not attempt to evaluate finished video.

Evidence, uncertainty and assumptions

Verified context: Kuku's assignment describes AI-generated 1-2 minute microdramas as an important production format. Runway's public prompting guide explicitly organizes video prompts around subject, action, camera and environment. **Unverified hypothesis:** artists may lose screenplay requirements while writing shot prompts, causing avoidable revisions. No Kuku artist interview, defect-rate measurement or savings study was conducted. Inputs below are original fictional test cases, not proprietary Kuku content.

Why Problem 3, rather than Problems 1 or 2?

Problem 1 requires ranking promos across many episodes; Problem 2 needs safety-critical legal grounding. Problem 3 is directly linked to Kuku's AI-video production and enables a narrow, independently checkable output: **is this shot prompt consistent with explicitly written story facts?** This is a scoped engineering choice, not a measured business priority comparison.

Proposed solution

Paste a screenplay excerpt and its video prompt; choose shot type and optional intent. The model extracts explicit actions, props, character/position facts and visible attributes, interprets paraphrases, then assigns **Supported / Contradicted / Missing / Ambiguous / Not Applicable**. The app shows exact text evidence, severity and minimal edits. A person approves or dismisses issues before any shot is generated.

System architecture

1 Streamlit input + shot scope	2 Replaceable Gemini adapter	3 Pydantic + Python verification
4 Evidence-backed review cards	5 Human decision + edited prompt	6 Unicode JSON / CSV exports

Worked example (human-authored expectation; NOT a Gemini prediction)

Screenplay	Arjun offers the red envelope to Meera.
Video prompt	Meera offers the red envelope to Arjun.
Expected finding	Contradicted / High: giver and recipient reversed. Quote both sources; change only the handover direction.

Sources: Kuku assignment, pp.1,3-5; Runway Text-to-Video Prompting Guide. These establish context, not the incidence of Kuku defects.

Implementation, evidence and limits

Technical implementation

Python 3.12 + Streamlit UI; Gemini Flash-family structured JSON behind a replaceable adapter; Pydantic schemas and deterministic evidence/ID/severity/count checks. One Gemini request performs three logical tasks: screenplay fact extraction, prompt interpretation, and shot-aware comparison. Quotes must exist verbatim in the user's inputs; failures remain errors, not fabricated results. Review state and raw model text are separate from final exports.

Actual evaluation - do not confuse software QA with Gemini accuracy

Evidence	Observed, 8 November 2026
Original labels	15 cases / 25 annotated facts; dataset validator: 0 errors
Keyword baseline only	Case-level TP 2, FP 3, TN 5, FN 5; precision 40.0%, recall 28.6%
Software and UI	187 pytest checks passed; localhost Streamlit health returned HTTP 200
Gemini semantic evaluation	0/15 labeled + 0/3 unseen cases run; no API key, no raw AI outputs or model accuracy

An observed failure, correction and retest

T06, real keyword-baseline false positive: the screenplay hides a key first, but the selected close-up shows only Nila's later reaction. A token-overlap checker demanded an off-shot action. **Change:** Gemini instructions now check shot timing/applicability before calling an omission. The instruction-contract regression passes; the frozen baseline still fails. **No authenticated Gemini failure or successful Gemini retest has been observed.** Software fixes also reject invisible/punctuation-only evidence and placeholder API keys; regressions pass.

Human control, limitations and scale

Each issue shows source evidence and a suggested minimal change. The artist chooses Accept, Dismiss as Intentional, or Needs Review; the editable corrected prompt and human choices never rewrite original AI findings. MVP is text-only, one shot at a time: no video-frame validation, continuity memory, measured quality uplift, Kuku field pilot or live LLM accuracy yet. For scale, batch independent shots with bounded concurrency, store reviews securely and audit disagreements against human labels.

Approximate running-cost method

Per shot = (input tokens x model input price + output/thinking tokens x output price) / 1,000,000, plus billed retries. **Per month** = shots reviewed x mean cost/shot. Record actual token usage, retries, latency and avoided regeneration cost in a pilot; none has been measured here. See official Gemini API pricing for current rates and tier conditions.

AI tools, prototype and required outputs

ChatGPT assisted product planning, Python/UI code, tests, documentation and this PDF; Gemini SDK is integrated **but was not invoked with credentials**; mocked-provider tests and a deterministic baseline are labeled as such. **Prototype:** complete runnable repository ZIP and offline demo included. **GitHub / Streamlit URL:** not published or verified. **Unedited raw Gemini outputs:** none exist - required for a fully complete submission. Deploy and run real cases before claiming them.